

Mathematics I

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The number system set of numbers

A **set** is a **collection of objects**. These are called **elements**.

Let n be an element, such as an integer number. The set of all positive integer numbers is called $\mathbb{N}$.

The statement: n is an element of $\mathbb{N}$ can be written as

$$n \in \mathbb{N}$$

The statement

$$x \in \mathbb{R}$$

indicates that x is an element of the set of real numbers $\mathbb{R}$.

Example of a **set**

Set of **Prime Numbers**:

$$P = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, \dots\}$$

Set of all positive integer numbers is

$$\mathbb{N} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, \dots\}$$

It is also possible to have sets as elements. For example

$$\{\{2, 1\}, \{3, 2\}, \{5, 3\}, \{7, 4\}, \{11, 5\}, \dots\}$$

where the 2nd number indicates that 2 is the 1st prime number,
3 is the 2nd prime number, ...

Mathematical notation

Have mathematical operations such as +, -, * and :

A statement such as

$$2 = 1 + 1$$

Indicates that *2 is the same as 1 more than 1.*

One should read: 2 is equal to 1 plus 1.

Similarly have

$$3 = 2 + 1 = 1 + 1 + 1$$

This can be continued without limit. The positive integers are also referred to as **natural numbers**. These number form $\mathbb{N}$.

If 0 is also included we shall call this set $\mathbb{N}^0$.

Mathematical operations

Let

$$x \in \mathbb{N}^0$$

then the following rules apply:

addition: $x + 0 = x$

subtraction: $x - 0 = x$

multiplication: $x \cdot 0 = 0 \cdot x = 0$

Division: $\frac{0}{x} = 0$

The following division is undefined: $\frac{x}{0}$

Negative Numbers

Calculations of the form

$$4 - 2 = 2$$

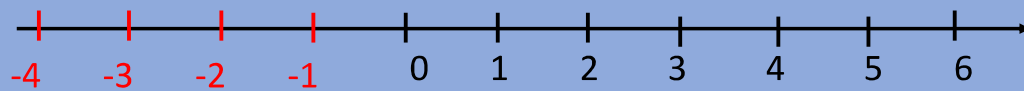
$$5 - 4 = 1$$

are possible within the natural number system. However, there are others such as

$$2 - 4 = -2$$

$$4 - 5 = -1$$

which require negative numbers. The negative numbers are simply an extension of the number line.



Axioms

Commutivity Law

Addition $a + b = b + a$

Subtraction $a - b = -b + a$ but $a - b \neq b - a$

Multiplication $a * b = b * a$

Associative Law

Addition $a + (b + c) = (a + b) + c$

Multiplication $a * (b * c) = (a * b) * c$

Subtraction $a - (b - c) \neq (a - b) - c$
but $a - (b - c) = (a - b) + c$

Distributive Law

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

Types of Numbers

Natural integer numbers

$$\mathbb{N}^0 = \mathbb{N}_0 = \{0, 1, 2, 3, 4, 5, \dots\}$$

An element of this set is referred to as $x_1 = 1, x_2 = 2, \dots$

The set of all **integer numbers** is

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

Rational numbers

$$2 = \frac{6}{3}$$

$$0.5 = \frac{2}{4}$$

Some rational numbers can be assigned an accurate value, but

note $\frac{5}{3} = 1.6666666 \dots$

Types of Numbers

Irrational numbers

An irrational number can not be expressed as a fraction of two integers.

Examples of irrational numbers

$$\sqrt{2} = 1.41421356 \dots$$

$$\phi = 1.61803398 \dots \text{ (golden ratio)}$$

$$e = 2.71828182 \dots$$

$$\pi = 3.14159265 \dots$$

Rational numbers and irrational numbers combine to form the **set of real numbers** $\mathbb{R}$.

Types of Numbers

Imaginary Numbers

Define imaginary number i such that $i^2 = -1$

Consider polynomial

$$f(x) = y = 5x^2 + 125$$

Solve the equation $f(x) = 0$.

Solution:

$$0 = 5x^2 + 125$$

Divide by 5 and obtain $0 = x^2 + 25 \Rightarrow x^2 = -25$

and obtain solution $x_1 = 5i$ and $x_2 = -5i$

Proof: $(x_1)^2 = (5i)^2 = 5^2 \cdot i^2 = -25$

$$(x_2)^2 = (-5i)^2 = (-5)^2 \cdot i^2 = 25 \cdot i^2 = -25$$

Visualization of complex Numbers

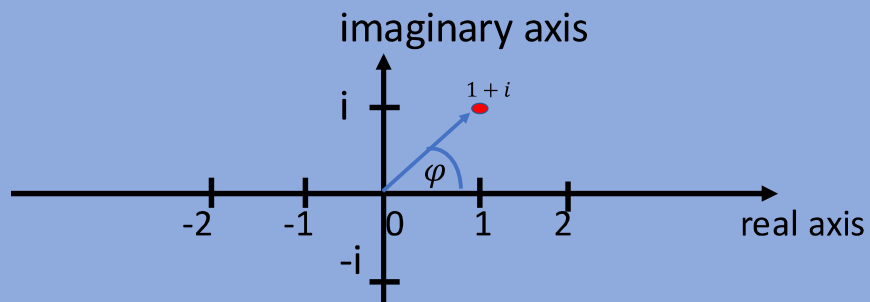
Take an complex number such as $x = 1 + i$.

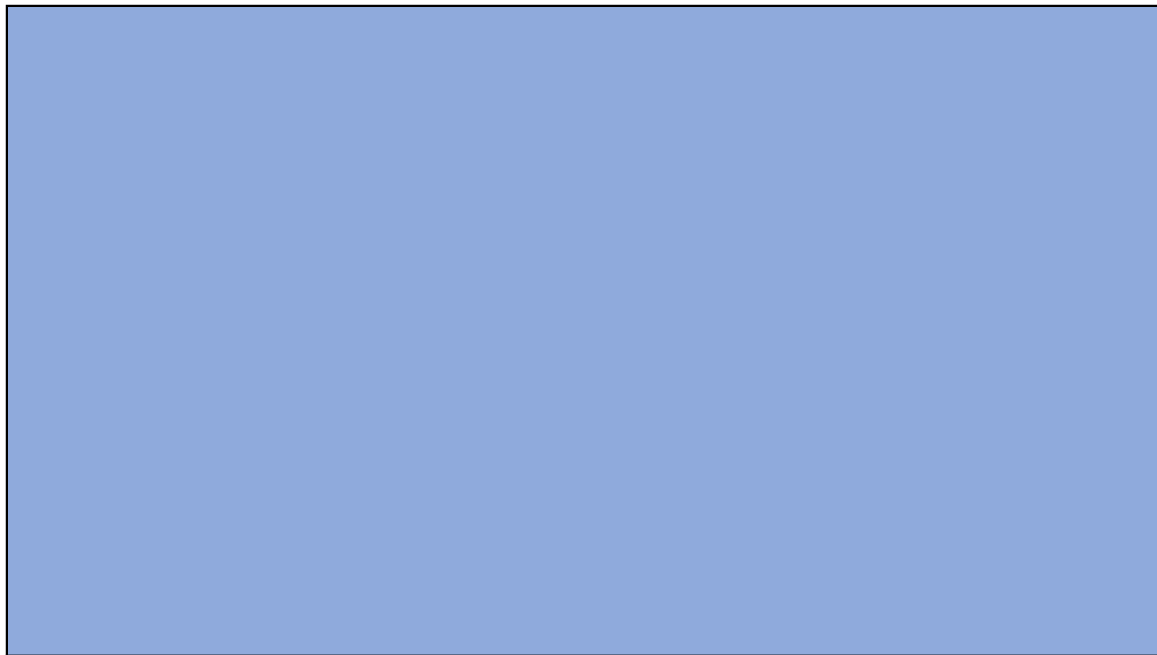
The complex number has a

Real part: $\operatorname{Re}(x) = \operatorname{Re}(1 + i) = 1$

Imaginary part: $\operatorname{Im}(x) = \operatorname{Im}(1 + i) = 1$

Both parts may be plotted as





Procedure:

Take the x-axis value to be the real part

Take the y-axis value to be the imaginary part

Identify the point in the x-y diagram

The distance of the point to the origin is the magnitude of the complex number.

$$|x| = \sqrt{x \cdot x^*} = \sqrt{(1 + i) \cdot (1 - i)} = \sqrt{2}$$

Here x^* is the **complex conjugate** which is defined as

$$x^* = (1 + i)^* = (1 - i)$$

Complex conjugation means: replace i by $-i$ in the complex number

Algebraic and transcendental numbers

Consider a polynomial of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x^1 + a_0$$

An example is

$$f(x) = y = 5x^2 + 3x - 1$$

$$f(x) = y = x^2 - 3x - \frac{7}{4}$$

All solutions x belong to the **set of algebraic numbers** $\mathbb{A}$.

Solve these equations, that is solve $f(x) = 0$.

Solutions:

$$\text{equ. 1: } y = 5x^2 + 3x - 1 = 0 \Rightarrow x_1 = -\frac{3}{10} + \frac{\sqrt{29}}{10} \quad x_2 = -\frac{3}{10} - \frac{\sqrt{29}}{10}$$

$$\text{equ. 2: } y = x^2 - 3x - \frac{7}{4} = 0 \Rightarrow x_1 = \frac{7}{2} \quad x_2 = -\frac{1}{2}$$

Alternative way of writing a complex number:

Rather than specifying the real and imaginary parts of a complex number one may also write it in the form

$$x = a + ib = r \cdot e^{i\varphi}$$

Here r is the magnitude of the complex number, given as,

$$r = \sqrt{a^2 + b^2}$$

and φ is the angle (see graph) between the real axis (x-axis) and the line from the origin to the point of the number in the complex plane. One has

$$\varphi = \arctan\left(\frac{b}{a}\right)$$

The function $e^{i\varphi}$ is the exponential function with complex argument.

Note that $|e^{i\varphi}|^2 = e^{i\varphi} \cdot e^{-i\varphi} = e^{i\varphi - i\varphi} = e^0 = 1$
for all values of φ .

Therefore $e^{i\varphi}$ must be located on a circle around the origin with radius 1.

One can show that the function can be written as

$$e^{i\varphi} = \cos(\varphi) + i \cdot \sin(\varphi)$$

Calculations with complex numbers:

Have two complex numbers x and y with $(a, b, c, d \in \mathbb{R})$

$$x = a + ib$$

$$y = c + id$$

Addition

$$x + y = (a + ib) + (c + id) = (a + c) + i(b + d)$$

Subtraction

$$x - y = (a + ib) - (c + id) = (a - c) + i(b - d)$$

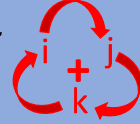
Multiplication

$$\begin{aligned} x \cdot y &= (a + ib) \cdot (c + id) = ac + iad + ibc + i^2 bd \\ &= (ac - bd) + i(ad + bc) \end{aligned}$$

The additional terms $+ijbg + ikbh + jicf + jkch + kidf + kjdg$ may be given the following meaning:

$$ij = k, \quad ji = -k, \quad jk = i, \quad kj = -i, \quad ki = j, \quad ik = -j$$

Then one obtains



$$\begin{aligned} xy &= (ae - bf - cg - dh) + \\ &\quad i(be + af) + j(ce + ag) + k(de + ah) + \\ &\quad + kbg - jbh - kcf + ich + jdf - idg = \\ &= (ae - bf - cg - dh) + i(be + af + ch - dg) \\ &\quad + j(ce + ag + df - bh) + k(de + ah + bg - cf) \end{aligned}$$

This is now again an object which has the same form as x and y . These entities are called **quaternions**.

Division

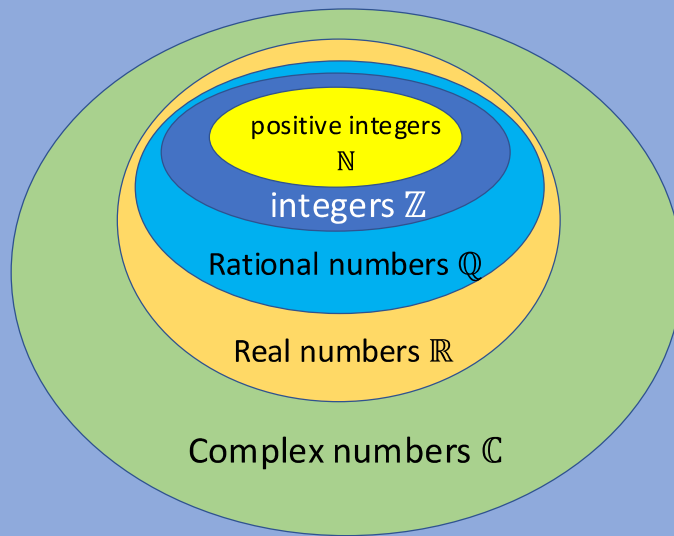
$$\frac{x}{y} = \frac{a + ib}{c + id}$$

This does not make sense in the traditional way. However, one may use

$$\begin{aligned}\frac{x}{y} &= \frac{x}{y} \cdot \frac{y^*}{y^*} = \frac{x \cdot y^*}{y \cdot y^*} = \frac{(a + ib) \cdot (c - id)}{c^2 + d^2} \\ &= \frac{(ac + bd) + i(bc - ad)}{c^2 + d^2}\end{aligned}$$

The number in the denominator is a real number and does not present a problem.

The number system may be visualized as follows:

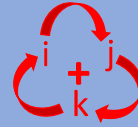


Quaternions

Define $x = a + ib + jc + kd$

with $a, b, c, d \in \mathbb{R}$ and $i, j, k \in \mathbb{I}$ $i^2 = j^2 = k^2 = -1$

A product of 2 numbers x and y results in



$$\begin{aligned}
 xy &= (a + ib + jc + kd) \cdot (e + if + jg + kh) = \\
 &= ae + ibe + jce + kde + iaf + i^2bf + jicf + kidf + \\
 &\quad + jag + ijb g + j^2cg + kjdg + kah + ikbh + jkch + k^2dh = \\
 &= (ae - bf - cg - dh) + i(be + af) + j(ce + ag) + k(de + ah) + \\
 &\quad + ijb g + ikbh + jicf + jkch + kidf + kjdg
 \end{aligned}$$